

IZZUL QUDSI

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PROFILE

Data geoscientist with 9+ years of experience in subsurface analysis, petrophysics, and exploration workflows. Currently pursuing a Ph.D. in Geology, applying artificial intelligence and hyperspectral data to advance digital subsurface characterization. Strong background in consultancy, R&D collaboration, and geoscience software development.

Core competencies: Subsurface interpretation • Subsurface data management • Python automation • AI for geoscience • Spectral Geology • Web-apps for geoscience

EXPERIENCE

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| 2025 | Student Co-op Training, National Petroleum Technology Company (NESR) <ul style="list-style-type: none">Supported R&D projects as data geoscientist.Contributed to the development of a prototype software platform for core data analysis. |
| 2021-Present | Group Leader, Geovarthia.id (non-commercial group discussion) <ul style="list-style-type: none">Led and organized the research, project and communication within the activities.Built web-app projects: Wellcast Geothermie Spectral-Box More output from this group is available online at https://www.geovarthia.id |
| 2019 - 2023 | Technical Geoscience Specialist, Horizon Perdana <ul style="list-style-type: none">Delivered consultancy support for well & seismic interpretation and petrophysics.Led the Data Management team to the initiation of the creation of the web application and its database infrastructure.Provided technical support for Elsiwave, OpendTect and Completion Manager. |
| 2014 – 2017 | Geoscientist, Trisakti University Oil and Gas Research Group (PUKESMIGAS) <ul style="list-style-type: none">Field geological mapping and core sample analysisWell and Seismic interpretation.Petrophysics analysis |
| 2013 – 2014 | MWD Engineer, Cougar Drilling Solution <ul style="list-style-type: none">Perform and record MWD survey and assist the Directional DrillerPerform tool inspection and check before and after rig-up rig-down |

TECHNICAL SKILLS

Programming/Data: Python (AI, ETL/QC, web backend), SQL, PPDM, APIs

Geoscience: Petrophysics, well log & core analysis, XRD/XRF, Hyperspectral, Seismic interpretation

Software: Petrel, Techlog, ArcGIS, ENVI, Oasis montaj, HampsonRussell, OpendTect

EDUCATION

King Fahd University of Petroleum and Minerals, Dhahran, Saudi Arabia

2023-now: Ph.D., *Geology* | GPA: 3.83/4 (ongoing)

- Winner of GEO4.0 Digitalization in Geoscience Symposium Hackathon

University of Twente, Enschede, The Netherlands

March 2019: Master of Science, *Applied Earth Science* | GPA: 7.9/10

- Faculty of ITC Excellence Scholarship awardee

Universitas Padjadjaran, Bandung, Indonesia

February 2013: Bachelor of Engineering, *Geology* | GPA: 3.34/4

PUBLICATIONS

- 2025 Reconstructing regional subsurface thermal structure of the Okavango rift zone, northern Botswana, and its low enthalpy geothermal potential. *Geothermics*. <https://doi.org/10.1016/j.geothermics.2025.103519>
- 2025 Leveraging Generative AI and Hyperspectral Drill-Core Imaging for Fully Automated Mineral Mapping of Unconventional Reservoir. *Applied Computing and Geosciences* <https://doi.org/10.1016/j.acags.2025.100298>
- 2025 Core Digitization: A Journey to Digital Transformation of Unconventional Mudstone Reservoir. SPE Conference at Oman Petroleum & Energy Show. <http://dx.doi.org/10.2118/224825-MS>
- 2022 Mineral Mapping on Hyperspectral Imageries Using Cohesion-based Self Merging Algorithm. *Jurnal Elektronika dan Telekomunikasi* <http://dx.doi.org/10.55981/jet.507>
- 2022 Geothermal Surface Manifestation Identification Using Airborne Hyperspectral Imagery. Case Study: Davis-Schrimp Geothermal Field, Salton Sea California at Indonesian Journal on Geoscience DOI: [10.17014/ijog.9.1.119-130](https://doi.org/10.17014/ijog.9.1.119-130)
- 2021 The Application of Parametric and Non-parametric Regression to Predict the Missing Well Log Data at Indonesian Journal on Geoscience DOI: [10.17014/ijog.8.3.385-399](https://doi.org/10.17014/ijog.8.3.385-399)

CONFERENCES

- 2025 Fast and Robust In-Situ Quantitative Mineral Analysis of Carbonate Rocks with AI-Driven Hyperspectral Unmixing Approach. MEOS-GEO 2025.
- 2024 Deep Learning-Assisted Quantitative Mineral Mapping of The Devonian Shale Core Using Hyperspectral Imaging. GEO4.0 Digitalization in Geoscience Symposium
- 2024 Estimating the Regional Thermal Structure of Crustal Thinning Zone in Northern Botswana at Topical Geothermal Workshop by SPWLA Saudi Arabia Chapter
- 2022 Mineral Identification on Hyperspectral Imagery of Rock Samples Using Machine Learning at 1st International Conference on Upstream Energy Technology and Digitalization – Universitas Pertamina, Jakarta. <https://doi.org/10.1063/5.0208704>
- 2021 Curie Point Depth Mapping using PyCurious from Aeromagnetic Data at GeoPython Conference 2021